

- (22) a. Expectations about specific content of cognitive effects
- b. Expectations about the kind of information that cognitive effects may convey
- c. Expectations about the relative size of sets of cognitive effects
- d. Expectations about the level of processing effort expected of the audience

Moreover, we have seen in table 1 that specific expectations of relevance are adopted on the basis of, and with the purpose of making concrete, the claim of optimal relevance which according to the communicative principle of relevance is communicated in every instance of ostensive communication.¹³ Unger (2006, pp. 107–142) points out that the inference processes involved in this step from the general claim of optimal relevance to the fine-tuning of relevance expectations may have profound impact on the processing of complex ostensive stimuli such as longer texts. However, the details of this sub-process of comprehension is rarely discussed in detail in the literature. The rationale for this is presumably due to the asymmetry of responsibilities of communicators and audience: whereas communicators must make sure that they choose utterances that are optimally relevant to their audience, audiences are licensed to presume that this is the case. This means that it is speakers who must evaluate what may be relevant to their hearers, whereas hearers do not need to double check what conclusions the speaker has come up with in this process.

Recall at this point that Rational Speech Act theory addresses both the speaker’s and the listener’s perspectives in a symmetrical way: the speaker reasons about how a ‘literal listener’ will understand an utterance under consideration, and how a ‘pragmatic listener’ will deviate from this. Likewise, the listener reasons about what a ‘literal speaker’ might have said to communicate a certain piece of information, and how a ‘pragmatic speaker’ deviates from this choice (see section 4.1.1). A corresponding probabilistic model of relevance theory capturing both communicator’s and audiences’ perspectives would need to capture the asymmetry of these communicative roles more clearly, making the description of the communicator’s inferences about the audience’s state of mind with respect to relevance much more complex. Clearly, the computational modelling of how communicators evaluate (and influence) what is relevant to other individuals is an important challenge to address. But in this paper we follow the path of classical relevance theory by concentrating the effort of explanation at the comprehension phase of communication.

The question arising at this point is how to handle relevance expectations in a probabilistic computational model. Expectations of relevance relating to cognitive effects can be expressed or paraphrased as questions (Unger 2006, pp. 143–173) which are often called *questions under discussion* (Beaver et al. 2017). These questions are about what may be inferred or concluded on the basis of information accessible in the context and provided in the utterance. In logic programming, inference is treated as answering the question whether a certain conclusion can be traced back to the ground facts provided by the knowledge base section of the program.¹⁴ In probabilistic logic programming, the answer to this question is not a binary choice between TRUE or FALSE, but a probability value indicating how likely it is that the queried proposition

¹³More precisely, in every instance of communication involving stimuli making a communicative and informative intention overt. There may be instances of ostensive behaviour that do not necessarily make a communicative intention overt, see [REF: find Sperber & Wilson 2024].

¹⁴See Flach (1994) for an accessible introduction to logic programming in Prolog. A fully interactive online edition can be found at the URL <https://book.simply-logical.space/src/simply-logical.html>.

is true. In the probabilistic programming language ProbLog (version 2)¹⁵, which we will be using in our implementation, inference processes answering the question as to how probable (i.e., how manifest) certain expected cognitive effects are, are triggered by using the `query()` predicate at the end of the program (see the code listings in the appendices).

Notice that this approach to checking relevance expectations by querying the program for answers to their corresponding questions first and foremost deals with specific relevance expectations as to content of cognitive effects (type 22a). By passing a whole set of queries about various questions under discussion we can also cover other types of relevance expectations of the types (22b) and (22c). These are the types of relevance expectations arising in our main example.¹⁶

5 Developing a probabilistic computational model of implicature comprehension

In this section we describe our implementation of a probabilistic computational model of implicature comprehension according to relevance theory in the probabilistic programming language ProbLog (Fierens et al. 2015).

5.1 Drawing a knowledge graph for encyclopaedic knowledge and context selection

Encyclopaedic knowledge linked to the concepts triggered by the words in the sentence conveyed in the utterance provides the starting point for context selection. In our example (1), we are dealing mainly with encyclopaedic knowledge relating to *coffee* and various other beverages that may typically be offered at parties, and their perceived properties (e.g. whether they block tiredness, enhance performance, etc.). This kind of knowledge can be effectively encoded in a graph structure. Inference over such a structure amounts to the query whether a path exists from a starting node A to a goal node B. ProbLog allows to encode probabilistic graphs, that is, the information that there is an edge (a link) between two nodes A and B can be annotated with a probability value (a number between 0 and 1). Figure 1 shows a schematic representation of the knowledge graph we use as a starting point. The thickness of the individual arrows linking nodes vary according to the probability values encoded. Lines 4–30 in the code listing in appendix B show the actual implementation of this graph.¹⁷

¹⁵See Fierens et al. (2015) for thorough and accessible introduction to probabilistic programming in ProbLog.

¹⁶We cannot directly model relevance expectations relating to processing effort alone, i.e. expectations of type (22d). Expectations of this kind are arguable raised in utterances communicating only a wide range of weak implicatures, as in example (21). Notice, however, that there is a close correspondence between relevance expectations of type (22c) and (22d): when the communicator promises the audience that they can satisfy their relevance expectations by investing a relatively high amount of processing effort, it follows that checking whether this promise is true amounts to processing a sizeable set of cognitive effects. This means that we can indirectly model relevance expectations of type (22d) by transforming them hypothetically as instances of (22c).

¹⁷For a more detailed discussion of the use of probabilistic graphs for concept and word meaning, see Erk and Herbelot (2023-11-30). Notice, however, that Erk and Herbelot tend to not making a clear distinction between word or concept meaning and encyclopaedic knowledge linked to concepts.

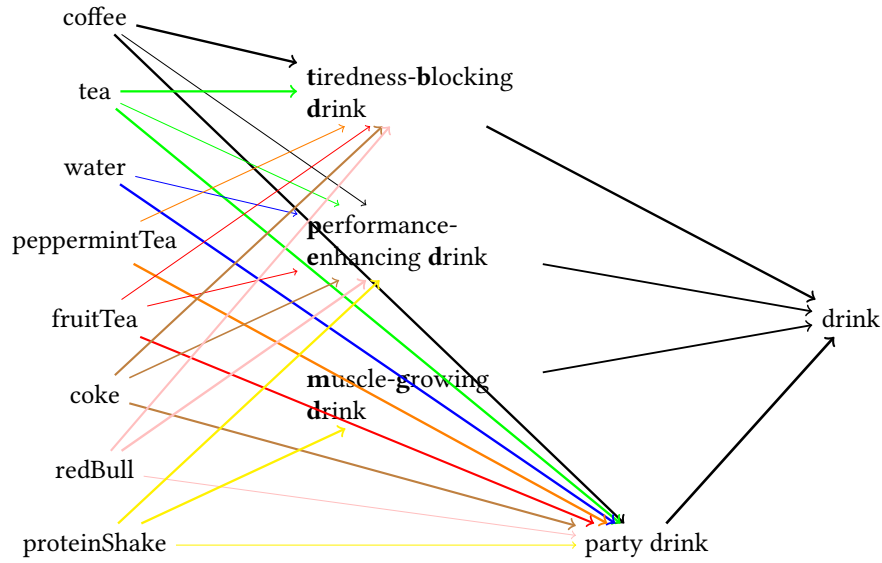


Figure 1: Graph representation of encyclopaedic knowledge in hypothesis 1

5.2 Linking words and concepts by inference

One task of utterance interpretation is the development of the logical form of the sentence uttered. This includes the question of which concepts are conveyed by words and phrases. In our example, a central question is what concept is conveyed by the compound word *energizing drink*. In other words, we need to establish a link between the word *energizing drink* and certain nodes in the knowledge graph representing encyclopaedic knowledge. We implement this by writing a disjunction of inference rules of the kind given in (23):

(23) X is an energizing drink, ...

- if there is a path from X to the concept TIREDNESS-BLOCKING DRINK, OR
- if there is a path from X to the concept PERFORMANCE-ENHANCING DRINK, OR
- if there is a path from X to the concept MUSCLE-GROWING DRINK.

Each disjunctive rule is assigned a probability value. See lines 40–45 in the code listing in appendix B.

These rules allow us to formulate queries of the kind *is coffee an energizing drink?* or *is tea an energizing drink?* and receive probabilistic answers. However, more interesting than begin able to execute such queries in isolation is the fact that we can use the probabilistic paths that such inferences open up in a net of wider inferences linking the description of Mary's utterance to its explicature, and thereby also to its implicatures.¹⁸ Figure 2 illustrates these links.

¹⁸In this way we approximate ad-hoc concept identification to disambiguation, i.e. a process of choosing among pre-existing concepts represented in the mind of the hearer. While ad-hoc concept resolution may in some cases come down to choosing among pre-existing concepts in this way, ad-hoc concepts are typically